

Bank Interest Program

Aim/Objective:

To develop a Java program to demonstrate method overriding and runtime polymorphism by finding the rate of interest for different banks.

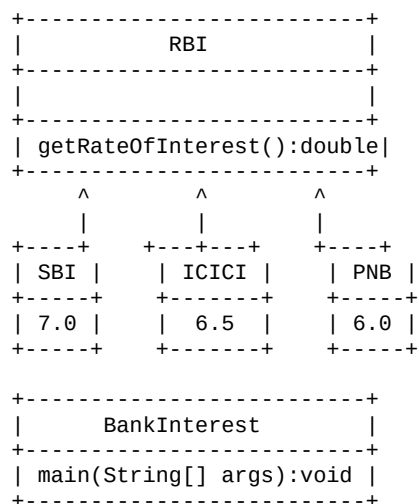
System Requirements:

- Operating System: Windows or any operating system supporting Java.
- Programming Language: Java.
- Java Development Kit (JDK).
- Text Editor or IDE: VS Code / Eclipse / IntelliJ IDEA.
- Command Prompt or Terminal.

Algorithm/Logic Description:

1. Start.
2. Create a base class named RBI with a method `getRateOfInterest()`.
3. Create SBI, ICICI, and PNB classes by extending RBI.
4. Override `getRateOfInterest()` in each bank class with its respective interest rate.
5. Read the bank name from the user.
6. Compare the entered bank name using `equalsIgnoreCase()`.
7. Create the corresponding bank object and assign it to an RBI reference variable.
8. If the bank name is invalid, display an invalid bank message and stop.
9. Call `getRateOfInterest()` using the RBI reference.
10. Display the interest rate of the selected bank.
11. Close the Scanner.
12. Stop.

UML Class Diagram:



Source Code:

```
import java.util.Scanner;

class RBI {
    double getRateOfInterest() {
        return 4.0;
    }
}
```

```

class SBI extends RBI {
    @Override
    double getRateOfInterest() {
        return 7.0;
    }
}

class ICICI extends RBI {
    @Override
    double getRateOfInterest() {
        return 6.5;
    }
}

class PNB extends RBI {
    @Override
    double getRateOfInterest() {
        return 6.0;
    }
}

public class BankInterest {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.println("Enter the Bank name to find the rate of Interest:");
        String bank = sc.nextLine();

        RBI rbi;

        if (bank.equalsIgnoreCase("RBI")) {
            rbi = new RBI();
        }
        else if (bank.equalsIgnoreCase("SBI")) {
            rbi = new SBI();
        }
        else if (bank.equalsIgnoreCase("ICICI")) {
            rbi = new ICICI();
        }
        else if (bank.equalsIgnoreCase("PNB")) {
            rbi = new PNB();
        }
        else {
            System.out.println("Invalid Bank Name");
            sc.close();
            return;
        }

        System.out.println("RBI rate of interest is : "
            + rbi.getRateOfInterest() + "%");

        sc.close();
    }
}

```

Compilation and Execution Commands:

Save the file as: BankInterest.java

Compile:

```
javac BankInterest.java
```

Run:

```
java BankInterest
```

Output Screenshot:

```
javac BankInterest.java  
java BankInterest  
Enter the Bank name to find the rate of Interest:  
SBI  
RBI rate of interest is : 7.0%
```

Result:

Thus, the Java program was successfully executed and method overriding with runtime polymorphism was demonstrated by displaying the rate of interest for the selected bank.